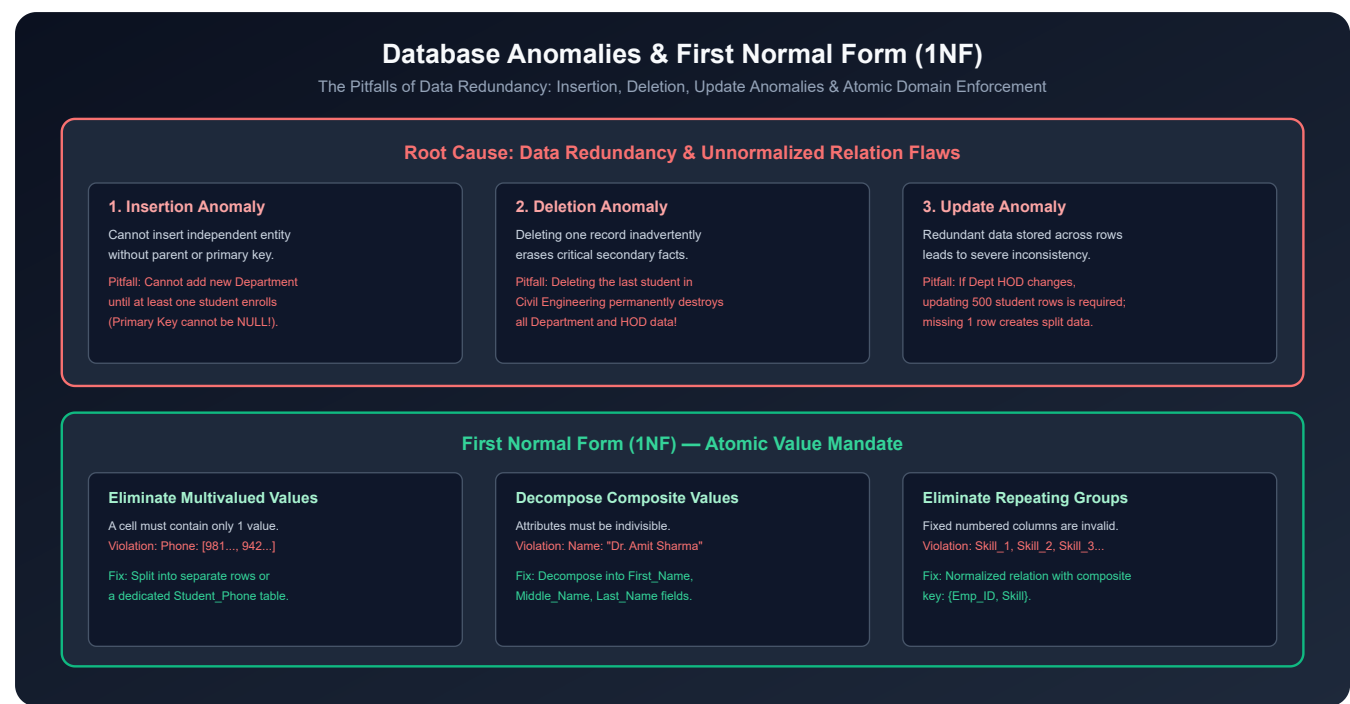


Module 05: Database Anomalies & 1NF



Maan lijiye hamare paas ek combined table hai:

STUDENT_DEPT(Roll_No, Name, Branch, HOD, HOD_Cabin)

Roll_No	Name	Branch	HOD	HOD_Cabin
101	Aman	CSE	Dr. Sharma	Block-A, 101
102	Priya	CSE	Dr. Sharma	Block-A, 101
103	Rohan	ECE	Dr. Verma	Block-B, 204
104	Sneha	ME	Dr. Gupta	Block-C, 301

2.1 Insertion Anomaly

Definition: Kisi nayi information ko database me add na kar paana jab tak kisi unrelated entity ka data available na ho.

Example: College me ek naya department shuru hua: AI_DS, jiske HOD Dr. Kapoor hain. Lekin abhi tak kisi student ne admission nahi liya.

Problem: Kyunki Roll_No Primary Key hai (aur Primary Key NULL nahi ho sakti by Entity Integrity Constraint), hum naye department ka record table me **insert hi nahi kar sakte!**

2.2 Deletion Anomaly

Definition: Kisi ek entity ka data delete karte waqt kisi important secondary entity ka data accidentally permanent delete ho jaana.

Example: Mechanical Engineering (ME) department me sirf ek student 104 - Sneha padh rahi thi. Sneha ne college chhod diya aur uska record delete kiya gaya.

Problem: Sneha ki row delete hote hi ME branch, uske HOD Dr. Gupta, aur cabin number ka saara record database se **hamesha ke liye gayab ho gaya!**

2.3 Update / Modification Anomaly

Definition: Redundant data ko update karte waqt agar sabhi copies update na hon, toh data inconsistency create hona.

Example: CSE ke HOD Dr. Sharma retire ho gaye aur nayi HOD Dr. Anita ban gayi.

Problem: Agar CSE me 500 students hain, toh 500 rows me jakar HOD change karna padega. Agar server crash ya network issue ki wajah se 490 rows update hui aur 10 reh gayi, toh database contradictory answer dega!

3. First Normal Form (1NF)

3.1 Formal Definition of 1NF

A relation schema R is in **First Normal Form (1NF)** if and only if the domain of each attribute contains only **atomic (indivisible) values**, and each attribute value in a tuple is a **single value** from that domain.

3.2 1NF Violations and Solutions

1. Multi-Valued Attributes

Violation: Kisi ek cell ke andar multiple values pack karna:

Roll_No: 101 | Name: Aman | Mobile: [9899112233, 9811223344]

1NF Solution: Multiple values ko alag-alag rows me expand karo:

- Row 1: (101, Aman, 9899112233)

- Row 2: (101, Aman, 9811223344)

- Or create a separate related table: STUDENT_PHONE(Roll_No, Phone).

2. Composite Attributes

Violation: Aise attributes jo sub-parts me divide ho sakte hain unhe single column me rakhna:

Address: "Flat 402, Royal Apts, Sector 62, Noida, UP, 201301"

1NF Solution: Break into atomic sub-components: House_No, Street, Sector, City, State, Pincode.

3. Repeating Groups / Numbered Columns

Violation: Ek hi type ke data ke liye predefined numbered columns banana:

STUDENT(Roll_No, Skill_1, Skill_2, Skill_3)

1NF Solution: Dynamic child table create karo: STUDENT_SKILLS(Roll_No, Skill).

4. Architectural Diagram

5. Summary Checklist: Is Relation in 1NF?

[x] Har column atomic values store karta hai? (No lists, no sets, no JSON arrays inside cells).

[x] Har row uniquely identifiable hai? (Primary Key identified).

[x] Columns ke names unique hain?

[x] Rows ka order meaningful nahi hai?